

Enhancing Quality Standards: A Case Study on the Zambian Bureau of Standards' Successful Implementation of Qualis LIMS

Overview

The Zambian Bureau of Standards (ZABS) is a statutory body under the Ministry of Commerce Trade and Industry of Zambia, established in 1982. It develops, publishes, maintains, and withdraws Zambian National Standards and related normative publications to meet Zambia's standardization needs. With the realignment of the National Quality Infrastructure (NQI) in 2018, ZABS focuses on supporting industry to implement standards that enhance product and service quality for industry growth and competitiveness.

ZABS' primary function is to test materials and products for conformity to set standards and good manufacturing practices. The testing services department operates based on international standards and is accredited to the ISO/IEC 17025 standard by SADCAS.

Operational Challenges

The laboratories at ZABS have been facing operational challenges due to their reliance on a hybrid system of manual documentation using excel spreadsheets along with an outdated customized legacy application. Using unproductive paper-based or legacy application led to increased costs, human errors during operations, difficulty in identifying and rectifying errors, low employee morale from repetitive tasks, issues with data integrity and security, and a lack of visibility of the status of work in progress.

To address these challenges, ZABS recognized the need to modernize and streamline its laboratory operations. They sought to implement a modern Laboratory Information Management System (LIMS) software to enhance efficiency, ensure data integrity, and establish a unified business process across all laboratories and departments.

Why ZABS chose Qualis LIMS

Being a statutory body, the main priorities of ZABS when it comes to selecting a LIMS solution was naturally security & compliance, with international standards & regulations. After conducting an extensive and rigorous search for the most suitable LIMS for their business processes, ZABS went with Qualis LIMS by Agaram Technologies. The decision to choose of Qualis LIMS was based on several key factors, such as:

- **Web-Based Application:** Qualis LIMS is a web-based application that can be accessed through standard web browsers. This allows for easy access from any computer device within the network, eliminating the need for software installation on individual systems and ensuring error-free installation and maintenance.
- **Scalability:** Qualis LIMS is an enterprise-scale application with high horizontal and vertical scalability. It can handle large user volumes by clustering web servers and deploying virtual servers, enabling smooth operation even with thousands of users simultaneously.
- **High Availability and Disaster Recovery:** Qualis LIMS can be configured for high availability, reducing the risk of non-availability. It can also be integrated with disaster recovery systems to meet specific criteria for individual customers, ensuring data and operational continuity.
- **Multi-Site Support:** Designed to accommodate organizations with multiple sites, departments, sections, and labs, Qualis LIMS allows for centralized management through a

single instance and database. It provides seamless coordination and data sharing across different locations and sections.

- **Standards Compliance:** Qualis LIMS adheres to W3C and ECMA scripting standards, widely recognized and utilized in the industry. This ensures compatibility and interoperability with other systems and reduces reliance on proprietary standards.
- **Security:** Qualis LIMS can be deployed as a secure application using HTTPS protocol and supports communication with servers through encryption of better than 128-bit. Additionally, specific tables containing confidential information can be encrypted within the database, enhancing overall data security.

By selecting Qualis LIMS, ZABS aimed to leverage its web-based architecture, scalability, multi-site support, standards compliance, and robust security features to enhance their laboratory operations and meet their specific requirements.

Improved & Standardised Processes

By implementing Qualis LIMS, ZABS significantly improved its laboratory operations by adopting industry best practices. The implementation process followed a structured workflow, comprising the following steps:

- I. **Planning:** ZABS collected and configured master data in the system, including organization hierarchy, user roles and rights, tests and specifications, and workflow setup.
- II. **Sample Registration:** Samples were collected, registered, labelled, stored, and divided into sub-samples. Tests were associated with each sample, enabling proper tracking and management throughout the process.
- III. **Work Allocation:** Tests were assigned to qualified users based on their competencies, ensuring that the right individuals were responsible for conducting the tests. Instrument maintenance & calibration was also implemented to assign tests to properly calibrated and maintained instruments.
- IV. **Analysis and Results Entry:** Analysts performed sample tests, entering the results directly into the system. Integration with Electronic Laboratory Notebooks (ELN) or Scientific Data Management Systems (SDMS) facilitated seamless automated data entry.
- V. **QA Verification and Approval:** Results were routed to the Quality Assurance (QA) department for verification. QA verifiers reviewed the results, approving, rejecting, or requesting recalculation/re-entry or re-testing as necessary.
- VI. **Report Generation:** The system generated comprehensive summary reports, including the Certificate of Analysis (COA). These reports were easily accessible within the system and could be dispatched to the required recipients.

By following the above standardized process with Qualis LIMS, ZABS achieved improved data accuracy, enhanced workflow management, and streamlined approval processes, ultimately leading to more efficient and reliable laboratory operations.

Other Benefits

In conclusion, implementing Qualis LIMS provided numerous benefits for ZABS:

- Significant time savings during implementation and roll-out.
- Operational efficiencies and cost savings through automation and elimination of paper-based processes and inefficient legacy applications.

- Reduced overhead costs and a low cost of ownership thanks to the standard technology architecture.
- Improved user experience with simplified operations and maintenance.
- Enhanced compliance with regulatory and industry standards, boosting customer confidence in data integrity and operational transparency.
- Minimized errors through automated processes, ensuring consistent and accurate result generation.

Overall, Qualis LIMS has brought about substantial improvements in ZABS' operations, delivering time and cost efficiencies, enhanced compliance, and a superior user experience.